# Structured Output Collapses Answer Diversity Across 44 Language Models

Tapan Parikh
Cornell Tech
tsp53@cornell.edu

July 2026

**Abstract**

When a language model must choose one answer from a large space of equally valid options, a format clause — *"Reply with JSON only"* — changes which answer it chooses. We re-run the One-Word Census [8], 31 category prompts with wide answer spaces asked of 44 models, with the reply requested in JSON: no schema enforcement, no constrained decoding, only the request. The field's convergence deepens sharply: on the unconstrained prompt ("Pick a word."), the modal answer's share rises from 40% to 63% and distinct answers fall from 53 to 28; mean answer-choice surprisal drops 1.80 to 1.58 bits. The tax is *progressive* and respects the instrument's resolution: seven of 44 models move individually (BH-FDR $q = .10$), all toward the mode, led by the panel's two most distinctive frontier models (the strongest explorer loses 1.32 bits — half its measured distinctiveness). The Llama-lineage models are the exception — both Meta-released Llamas and a community merge *diverge* under JSON where the field compresses, a descriptive lineage link we flag but do not test. The format acts as a sharpener, not a re-indexer — the plain-chat modal answer survives in 28 of 31 categories, and the three flips occur exactly where the plain mode was weakest. The column also shows that a model's *defaults are register-indexed*: a within-run re-sample ($n$=20) finds JSON significantly shifts 53% of a model's stable chat defaults — mostly back to the crowd's answer — and *installs* defaults that never appear in chat (Claude Fable 5 answers *cerulean* for colour 0% of the time in chat, 100% in JSON). Personality is indexed to the channel, as the census found answers indexed to wording. Full-battery controls reveal a register gradient: the compression is significant and specific to the answer-delivery formats models are trained to speak (JSON −0.22 bits, $p = 10^{-4}$; XML −0.18, $p = .003$; sign-flip permutation over models, corroborated over categories), absent for YAML and CSV ($p > .15$), and *reversed* for an arbitrary non-data wrapper — square brackets significantly loosen the field (+0.16, $p = .002$). On the unconstrained prompt, all four serializations concentrate the pool (serendipity 41% plain → 52–65%), so a corpus-register component remains visible; but the battery-wide gradient weighs toward tool-use post-training as the dominant mechanism. Enforcing the schema at the decoder (`response_format`) compresses no further than the request (−0.03 bits), locating the collapse in the model's response to the register rather than the decoder. Structured output is how software consumes language models; that surface is served by a measurably more homogeneous model than the chat surface on which models are evaluated, compared, and chosen.

# 1 Introduction

Software does not consume language models in prose. It asks for JSON. Every agent that calls a tool, every pipeline that extracts a field, classifies a record, fills a schema, or routes a request receives the model's answer inside a data structure — and that traffic increasingly dwarfs the chat window in which models are benchmarked, ranked, and chosen. This paper asks a question about that surface: when the same model answers the same question in a serialization format instead of prose, does it give the same answer?

The companion One-Word Census [8] built a deliberately narrow instrument. When a model must choose one answer from a large space of equally valid options — *"Name a tree. Reply with one word only."* — which does it choose, and how often is it the same answer the rest of the field chooses? The metric is *answer-choice surprisal*: a leave-one-out measure, in bits, of how unlikely a model's answers are under the pooled answers of every other model, computed by exact match on normalized one-word replies — no embeddings, no LLM judge, no human annotation, roughly a dollar per model. Across 44 models it found a monoculture (*oak* takes 94% of tree answers; *serendipity* 41% of answers to an unconstrained "Pick a word.") structured by a wide per-model conformity range and, for a handful of models, stable off-modal defaults that constitute a measurable character.

We re-run that instrument unchanged in every respect but one: the reply is requested in a serialization format. *"Name a tree. Reply with JSON only, in the form {"word": "<your answer>"}."* No schema enforcement, no constrained decoding, no token masking — only the request. The 31 prompts, the 44-model panel, and the conditions (no system prompt, requested temperature 1.0, four samples) are identical to the census; the single manipulated variable is the register in which the answer is delivered. Because surprisal is computed *within* each format column — JSON answers scored against the JSON field — a column's convergence is an internal property of that column, not an artifact of comparing one register against another.

The field converges harder, and it does so unevenly. Our contributions:

1. **A progressive format tax** (§4.1). Field-mean surprisal falls from 1.80 to 1.58 bits ($-0.22$ per model, $p = 10^{-4}$), but the drop is not uniform: seven of 44 models move individually (BH-FDR $q = .10$), all toward the mode, led by the panel's most distinctive models (the strongest explorer loses 1.32 bits, half its measured distinctiveness), while the conformist floor is immobile. The remaining deltas form a noise plateau we decline to rank.

2. **A sharpener, not a re-indexer** (§4.2). The plain-chat modal answer survives in 28 of 31 categories; the three flips occur exactly where the plain mode was weakest, and land on answers that read as the example-payload canon of documentation.

3. **Register-indexed defaults** (§4.4). A within-run re-sample ($n$=20) shows the register significantly shifts 53% of a model's stable chat defaults (mostly back to the crowd) and *installs* register-only defaults absent from chat — read from discrete answer behavior, not noisy score deltas.

4. **A register gradient** (§4.5). Compression is significant and specific to the answer-delivery formats models are trained to speak (JSON, XML), absent for YAML and CSV, and *reversed* for an arbitrary non-data wrapper — weighing the mechanism toward tool-use post-training while leaving a corpus-register component visible.

5. **A lineage-linked exception** (§4.5). The Llama-lineage models *diverge* under JSON where the field compresses; with few members and undisclosed base weights we report this descriptively, as a lead rather than a tested effect.

6. **A cheap, reusable method**. The manipulation is one extra column on a frozen instrument, and format compliance falls out of every run as a standing capability metric.

# 2 Related work

**Format restrictions and accuracy.** The nearest prior line asks whether format *restrictions* degrade task performance. Tam et al. [10] report that requiring structured output lowers reasoning accuracy; Kurt [7] contest the result, attributing it to prompt design and parsing rather than the format itself. Our question is orthogonal to this debate: our prompts have no wrong answers, so there is no accuracy to degrade. We measure which valid answer is chosen, not whether a correct one is reached.

**Decoder-level enforcement.** A 2026 line names the cost of *enforcing* structure at the decoder — the "constraint tax" [2, 3] and "format tax" [4]: grammar-constrained decoding raises validity while lowering correctness and suppressing behaviors such as tool calls. But they constrain the sampler with token masking; we mask nothing and only ask. The effect we measure therefore lives in the model's response to a register, not in a decoding constraint.

**Diversity within grammars.** aut [1] engineer diversity back into constrained generation with automata-based steering. That work presupposes exactly the collapse we characterize — it is an engineering fix for a phenomenon the present paper measures at the level of a mere request, with no grammar in the loop.

**Surface form and register.** Sclar et al. [9] show model accuracy is sensitive to spurious formatting choices; the census showed answer *choice* is sensitive to wording (*gemstone* vs. *precious stone*). This paper adds a third axis: not how the question is worded but the register in which the answer is requested. Underlying all of this is the mode-collapse literature [5, 6, 11], which documents the convergence we exploit and notes that LLM judges systematically prefer modal outputs — one reason our instrument stays judge-free.

# 3 Method

## 3.1 Instrument

The instrument is the *one-word census* unchanged [8]: 31 single-turn prompts (30 category prompts of the form *"Name a[n] X"* plus one unconstrained *"Pick a word"*), no system prompt, requested temperature 1.0, four samples per cell. Each answer is scored by its add-one-smoothed leave-one-out answer-choice surprisal against the pooled answers of the other 43 models, and a model's headline score is the mean over its valid answers, in bits. Replies are reduced to a single normalized token by exact-match rules and a mechanical junk guard (§3.3).

Surprisal is computed *within* each format column. JSON answers are scored against the JSON field, plain answers against the plain field. A column's convergence is therefore an internal property of that column — it is never a comparison of JSON text against a chat reference, and so cannot be an artifact of the register shift itself. The per-model Δ-*surprisal* (JSON minus plain) is the headline quantity: how many bits of a model's distinctiveness the register removes.

### 3.2 Format columns

The plain-chat column is the census transcripts. Each of five format columns re-runs the full 31-category battery for all 44 models at four samples per cell (5,456 calls per column): JSON, XML, YAML, CSV, and a non-data "brackets" wrapper, the last included to separate serialization from mere structure. The clauses are appended verbatim to the census prompt:

- **json**: Reply with JSON only, in the form {"word": "<your answer>"}.
- **xml**: Reply with XML only, in the form <word>your answer</word>.
- **yaml**: Reply with YAML only, in the form ‘word: <your answer>‘.
- **csv**: Reply with CSV only: a header row ‘word‘, then one data row with your answer.
- **brackets**: Reply with your answer inside square brackets only, like [answer].

No `response_format` parameter, no tool schema, and no constrained decoding is used anywhere: every column is an ordinary chat completion whose only difference from the census is the appended clause. This is a deliberate scope choice: these columns measure the effect of the *request*. An engine-enforced `response_format` column is analyzed separately in §4.6.

### 3.3 Parsing and hygiene

Each reply is reduced to a single answer token in two steps. First the format wrapper is stripped by a per-format regular expression — the contents of the JSON `"word"` field, the text between <word> tags, and so on — and the extracted string is then passed through the census normalization and junk guard unchanged. A reply that does not match its format's wrapper falls back to the census rule applied to the raw text, so that a malformed reply cannot smuggle in a spurious "novel" answer: the last word of a broken-JSON essay is scored exactly as the census would score it. Parse-plus-junk survival is at least 99% per model per column, and we audited per-model survival for every model whose surprisal *rose*, since non-compliance is the one artifact that manufactures apparent divergence (§4.5).

Of the 27,280 format cells, 91 (0.3%) are unrecoverable nulls, concentrated in YAML (56) and negligible in JSON (1); all distributional claims are computed over the surviving cells, with the per-column parse survival audited above.

## 4 Results

### 4.1 The format tax is progressive

Field-mean answer-choice surprisal falls from 1.80 bits in plain chat to 1.58 in JSON. Averaged over the 44 models the per-model Δ is −0.22 bits, significant on both exchangeable units (§4.5);

but the mean is the least informative summary available, because the effect is concentrated rather than spread. The shape is a divergent tail collapsing onto a fixed floor (Figure 1).

The models that lose the most are the ones that had the most to lose. DEEPSEEK-V3.2, the census's one genuine explorer among frontier-lab models, falls from 2.63 to 1.32 bits (−1.32) — half its measured distinctiveness — the moment the answer is requested as JSON. HERMES-4 loses 0.98, GPT-4O-MINI 0.92, GPT-4-TURBO 0.87, GPT-5.6-SOL 0.66. The conformist floor barely moves: CLAUDE-OPUS-4.8 −0.17, GROK-4.5 −0.07, CLAUDE-SONNET-5 −0.05. There is nowhere below the floor to go, and the tail falls toward it, though the raw range is nearly unchanged (2.2 to 2.0 bits): a Llama-derived minority *diverges* under JSON (§4.5) and holds up the top, so the compression is in the mean and the collapsing tail, not the extremes. On the unconstrained "Pick a word" prompt the same collapse appears with no category to anchor it: *serendipity* rises from 40% of the pool to 63% and the number of distinct words falls from 53 to 28.

Which individual movements are real we take up in §4.5; the progressive *shape* — tail down, floor fixed — is a property of the aggregate and does not depend on ranking the middle of the table.

## 4.2 A sharpener, not a re-indexer

If the register merely added noise, or merely imposed structure, it could move probability anywhere. Instead it sharpens the distribution the model already had. In 28 of 31 categories the JSON modal answer is the same word as the plain-chat mode, and less than 4% of JSON answer-mass lands on words the plain census never produced. The register moves mass *onto the existing mode*, not onto new answers.

The three exceptions are diagnostic: each is a category whose plain-chat mode was weak, and in each the JSON winner is a documentation-flavored prototype. Insect flips from *ant* (43%) to *butterfly* (60%), board game from *chess* to *monopoly*, dance from *salsa* to *tango*. *Butterfly*, *monopoly*, and *tango* are the example-payload canon of API tutorials — the strings that fill placeholder fields in documentation. The register amplifies its own conditional prior, and that prior wins only where the chat prototype was too weak to hold. This mirrors a contrast from the census: a *semantic* reframe re-indexes the distribution onto a new mode, while a *channel* reframe like this one sharpens the peak the model already had.

## 4.3 The register moves the mode, not the temperature

A natural worry is that we are measuring a sampling change rather than a change in preference — that some providers quietly cool the sampler when a request looks like structured output. The census's self-distinctness statistic (distinct answers divided by samples, within model and category) is the effective-temperature proxy that settles it. At the field level it is nearly flat across all six columns — plain 0.42, JSON 0.39, XML 0.40, YAML 0.42, CSV 0.43, brackets 0.43 — while surprisal drops 0.22 bits under JSON. A serving-layer temperature cut would crater self-distinctness wherever it cut; it does not. The collapse is *positional*: mass relocates onto the field's mode without the model sampling any less around it.

The within-model narrowing that does exist is progressive, on the same models as the surprisal effect: HERMES 0.71 → 0.52, DEEPSEEK-V3.2 0.56 → 0.42, SONAR 0.50 → 0.40, HAIKU-4.5 0.42 → 0.33, while the conformist floor is flat (SONNET-5 0.30 → 0.27) and CLAUDE-FABLE-5 holds at 0.32 in both registers — its steadiness is conviction, not a cold sampler. The tax is thus progressive on both of the census's axes, and the dissociation between a large surprisal move

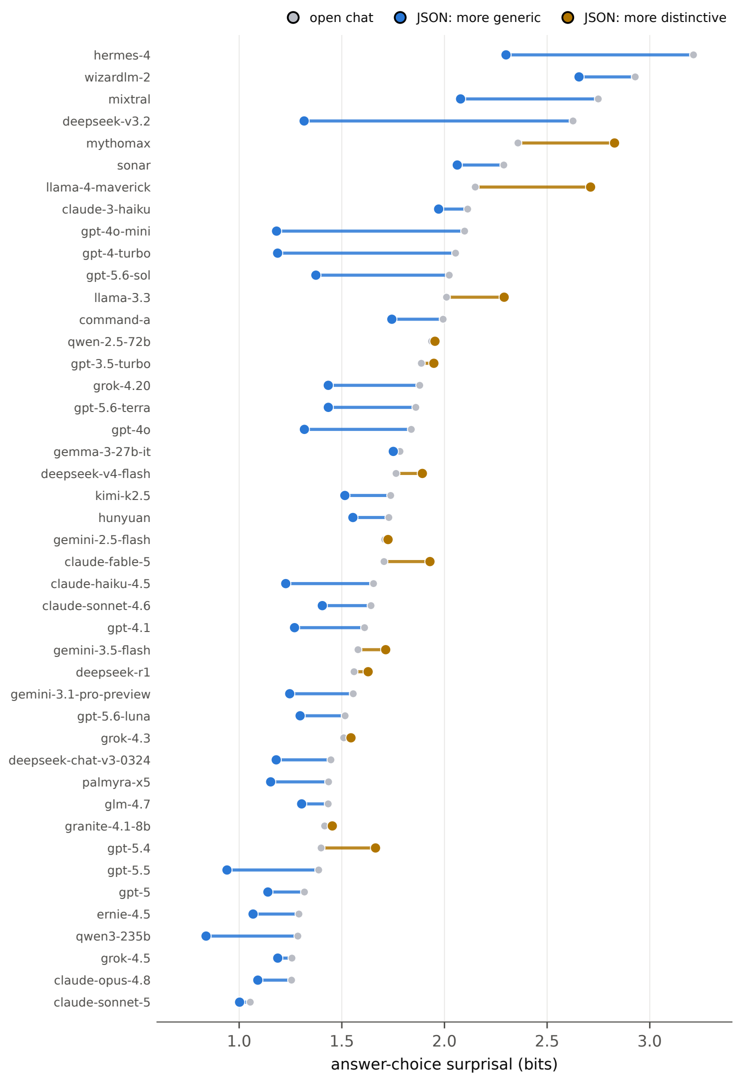


**Figure 1:** Answer-choice surprisal for all 44 models, open chat (gray dot) versus JSON (colored dot), sorted by the open-chat score. Blue marks a model that becomes *more generic* under JSON (31 of 44); amber marks one that *diverges* (13 — the Llama-derived group and a few others, §4.5). The distinctive models at the top slide farthest toward the conformist floor, which is itself immobile, and the field mean falls from 1.80 to 1.58 bits. The raw range is nearly unchanged (2.2 → 2.0) because the amber divergers hold up the top: the compression is in the mean and the tail, not the extremes.

and a small self-distinctness move is itself the signature of a reshaped conditional distribution rather than a rescaled one.

## 4.4 Defaults are register-indexed

The sharpest evidence that personality is register-dependent comes not from surprisal scores, which are noisy for individual models (§4.5), but from discrete answer behavior. Call a model's *stable default* in a category an off-modal answer it gives repeatedly — a standing refusal of the field's first choice, like FABLE's *gouda* for cheese. The four-sample census flags 144 such defaults across the panel. Four samples, however, cannot tell a genuine default from a lucky streak, nor a register effect from ordinary run-to-run drift, so we re-sampled every default cell at $n = 20$ in *both* registers within a single run. This settles three questions the four-sample data could not.

**The defaults are real, and the register erodes them.** At $n = 20$ the flagged defaults are genuine (median per-sample probability 0.90), not sampling artifacts. Requesting JSON then *significantly* shifts the answer distribution for 76 of 144 (53%) of them (Fisher exact per cell, Benjamini–Hochberg $q = .10$), against a false-positive floor of $\sim$10%, and the shift is directional: 29% revert outright to the field's modal answer. The register pulls a model's idiosyncratic default back toward the crowd.

**The register also installs defaults.** The move is not only subtractive. Of the JSON answers a model gives four-of-four but *never* produces in chat, 81% survive the $n = 20$ re-sample as genuine register-only defaults: FABLE says *cerulean* for colour 0% of the time in chat and 100% in JSON ($p \approx 10^{-11}$), and *carpenter* for occupation 0% → 90% ($p \approx 10^{-9}$); OPUS-4.8 acquires *phoenix*, SONNET-4.6 *gold*. Some of the field-level mode flips of §4.2 are these acquisitions in aggregate — one model's JSON-only *butterfly* (15% → 100%) or *monopoly* (5% → 100%).

**Same default, opposite response.** Because the register both erases and installs, models that look identical in chat diverge under it. GPT-5.6-TERRA and FABLE both answer *mango* for fruit, twenty runs of twenty in chat; under JSON TERRA flips to *apple* (20/20, $p \approx 10^{-11}$) while FABLE holds *mango* (19/20). The right object is therefore not one personality the register reveals or hides but a *per-register defaults profile*: a model's character is indexed to the channel it is asked through, the same way the census found answers indexed to the wording of the question.

## 4.5 Serialization vs. structure: the register gradient

**The gradient.** Extending the same battery to four further formats separates *serialization* from mere *structure*, and the result is a gradient rather than a uniform effect. Conditioning on compliance — restricting each format to the models that produce its wrapper at least 90% of the time, so incapacity stays out of the distributional signal — the field-mean $\Delta$-surprisal is $-0.22$ bits for JSON, $-0.22$ for XML, $-0.08$ for YAML, $-0.02$ for CSV, and $+0.16$ for brackets. The two formats that compress the field are the two models are trained to *answer* in: JSON and XML, the registers of function calls, tool use, and structured-output modes. The data formats models mostly only *read* — YAML and CSV — do not reliably compress it, and an arbitrary non-data wrapper (square brackets) *loosens* it.

**Significance.** We test both exchangeable units with a sign-flip permutation (20,000 draws): field entropy paired over the 31 categories, and within-column surprisal paired over the 44 models. JSON and XML compress on both units (entropy $-0.20$, $p = .0006$ each; per model $-0.22$, $p = 10^{-4}$ and $-0.18$, $p = .003$). YAML and CSV are not significant on either unit ($p = .76$ and .15 by category; .81 and .16 by model). Brackets significantly *loosens* on both (+0.14 entropy, $p = .005$; +0.16 per model, $p = .002$). The JSON-versus-YAML difference, paired by category, is itself significant ($p < 10^{-4}$), and DEEPSEEK-V3.2's individual collapse clears $p < 10^{-4}$ alone. The compression is thus specific to the trained answer-delivery formats, which moves the weight of the mechanism toward tool-use post-training rather than a generic property of serialized text.

**But not post-training alone.** The corpus-register account is not dead, because on the unconstrained "Pick a word" prompt *every* serialization concentrates the pool, including the two with no net battery effect: *serendipity* rises from 41% in plain chat to 52% (CSV), 59% (XML), 64% (JSON), and 65% (YAML), while brackets holds at 38%. YAML's single-category concentration is real; it simply washes out across the full battery, which is why its net effect is null. A black-box study cannot fully separate what the training corpus taught — that text inside a data structure is the canonical value — from what tool-use tuning reflexively reinforced; our reading is that both are present and the post-training component dominates the battery-wide gradient.

**Two companion phenomena.** Compliance is not a nuisance to discard but a control that must be applied. *Format incompetence* masquerades as divergence: GRANITE and MYTHOMAX emit a valid CSV wrapper only about 1% of the time, and their unwrapped replies, scored naively, read as high surprisal — GRANITE shows a spurious +1.69 bits in CSV before conditioning. This is why every distributional figure above is compliance-conditioned. Distinct from incompetence, there is a genuine *register-diverger*: LLAMA-4-MAVERICK is 100% compliant in JSON and CSV and still moves *away* from the field in every format (+0.4 to +0.7 bits). Register reaction is heterogeneous, not uniformly compressive.

**Per-model deltas: three tiers.** Applying the census's tiers-not-ranks discipline to the deltas: (a) exactly seven of 44 models have an individually significant $\Delta$ at BH-FDR $q = .10$, and all seven are compressions — DEEPSEEK-V3.2 $-1.31$, HERMES $-0.96$, GPT-4O-MINI $-0.92$, GPT-4-TURBO $-0.87$, GPT-5.6-SOL $-0.65$, QWEN3 $-0.45$, GEMINI-3.1-PRO $-0.31$. (b) The remaining $\sim$37 models — including every model with a positive $\Delta$ — are individually indistinguishable from zero, and their mid-table ordering carries no information; we do not interpret it. (c) One effect survives only at family altitude.

**A family-level response.** The models with the largest positive deltas share a lineage: the two Meta-released Llamas, LLAMA-4-MAVERICK (+0.56) and LLAMA-3.3 (+0.28), both diverge under JSON, as does a community Llama-2 merge (MYTHOMAX, +0.47). We name the group by provenance rather than by outcome, but with so few members this is a descriptive observation, not a tested effect, and we quote no aggregate $p$-value. That a community merge sharing the base but none of Meta's post-training also diverges points at the base weights over any lab's tuning — yet those weights are undisclosed, so we flag the lineage link as a lead, not a result.

### 4.6 Enforcement adds little the request did not

A natural objection is that the collapse is a decoding artifact. It is not: enforcing the schema at the decoder compresses no further than asking for it. We ran one more column — the JSON clause plus a strict `response_format` json_schema — on the 36 of 44 models whose providers support it (the other eight return no valid output, a feature-acquisition signal in itself). Field-mean surprisal is 1.57 bits under the request and 1.54 under enforcement, against 1.79 in open chat: the request does the −0.22-bit work and enforcement adds −0.03 more. The effect lives in the model's response to the register, not in the sampler. Per-model reactions are heterogeneous and confounded — `response_format` is a native decoder constraint for some providers and a gateway coercion for others — so we read only the battery-wide mean and release the data for a native-only replication.

## 5 Limitations

Several limitations bound these claims. The data are a single snapshot served through one channel (OpenRouter), and requested temperature 1.0 is not honored uniformly across providers — which is why we report self-distinctness alongside surprisal and rest the personality claims on discrete four-of-four behavior. Surprisal is panel-relative: a model's bits depend on the field it is scored against, so the absolute numbers are not portable across panels, although the within-column and $\Delta$ comparisons are internal to this one. Each format is probed with a *single* clause wording, so we cannot separate the register from the particular phrasing that invokes it; a clause-paraphrase column is the natural control. Finally, we vary the register through prompt-level *requests* and one enforced `response_format` counterpart (§4.6); tool-call framing and other structured-output pathways remain unmeasured, and enforcement is itself served heterogeneously across providers, so a provider-controlled native-enforcement replication would sharpen the per-model picture.

## 6 Discussion

The result has one blunt practical consequence. Models are evaluated, compared, ranked, and felt out in chat — leaderboards, vibe checks, and human preference are all collected on the prose surface — but software consumes them through structured output, and that surface is measurably more collapsed. Every diversity number the census reported is, for the deployment tier that increasingly matters, an overstatement: the model an agent consults has narrower tastes than the one a user chatted with, and the gap is a fixed cost of the register rather than a tail risk. Synthetic-survey, LLM-judge, and tool-selecting pipelines all read the JSON persona.

The register is also a cheap instrument in its own right. One extra column on a frozen battery dissociates register-dependent divergence (a costume) from register-independent conviction (a default that holds in both channels), and format compliance falls out for free as a generational capability track — older models cannot speak some of these registers at all, and that acquisition has a date a re-run will record. If the battery-wide compression we attribute mostly to tool-use post-training is being trained in per release, a fixed public instrument run on every model is what would detect it — the same argument the census makes for measuring conformity over time, now pointed at the register in which conformity is worst. The immediate next columns are a provider-controlled native-enforcement replication and tool-call framing (§4.6), clause paraphrases, and an embedded variant that reads choices out of prose through regex-addressable slots — never LLM judges, which prefer the mode.

## Data and code availability

All code and data are in the `studies/structured/` directory of the modelun repository: `run_formats.py` (the battery runner), `analyze.py` (the metric, junk guard, and Δ-surprisal), `probe_significance.py` (the permutation tests), and `probes/format_register.json` (the raw replies for all six columns). The plain-chat baseline is the census transcripts; this study scores against the 44-model panel. An interactive explorer — scorecard, per-category and per-format tables, per-model drill-downs, and a compliance tab — is published alongside.

## Note on AI usage

This work was done in collaboration with Claude (Opus 4.8 and Fable 5), which helped run the battery, build the analysis, and draft the text; the research questions and interpration are the author's. Both models are also subjects of the study.

## References


[1] Automata-based steering of large language models for diverse structured generation, 2025. arXiv:2511.11018.

[2] Constraint tax in open-weight llms: An empirical study of tool calling suppression under structured output constraints, 2026. arXiv:2606.25605.

[3] The constraint tax: Measuring validity-correctness tradeoffs in structured outputs for small language models, 2026. arXiv:2605.26128.

[4] The format tax, 2026. arXiv:2604.03616.

[5] Liwei Jiang, Yuanjun Chai, Margaret Li, Mickel Liu, Raymond Fok, Nouha Dziri, Yulia Tsvetkov, Maarten Sap, Alon Albalak, and Yejin Choi. Artificial Hivemind: The open-ended homogeneity of language models (and beyond). *Advances in Neural Information Processing Systems (NeurIPS)*, 2025. arXiv:2510.22954.

[6] Robert Kirk, Ishita Mediratta, Christoforos Nalmpantis, et al. Understanding the effects of RLHF on LLM generalisation and diversity. In *International Conference on Learning Representations (ICLR)*, 2024. arXiv:2310.06452.

[7] Will Kurt. Say what you mean: A response to "let me speak freely". https://blog.dottxt.ai/say-what-you-mean.html, 2024.

[8] Tapan Parikh. The one-word census: Answer-choice conformity across 44 language models, 2026. arXiv:2607.12796; data and explorer at https://github.com/tap2k/modelun.

[9] Melanie Sclar, Yejin Choi, Yulia Tsvetkov, and Alane Suhr. Quantifying language models' sensitivity to spurious features in prompt design, or: How i learned to start worrying about prompt formatting. In *ICLR*, 2024. arXiv:2310.11324.

[10] Zhi Rui Tam, Cheng-Kuang Wu, Yi-Lin Tsai, Chieh-Yen Lin, Hung-yi Lee, and Yun-Nung Chen. Let me speak freely? a study on the impact of format restrictions on performance of large language models. In *Proceedings of EMNLP 2024: Industry Track*, 2024. arXiv:2408.02442.

[11] Yiming Zhang, Harshita Diddee, Susan Holm, Hanchen Liu, Xinyue Liu, Vinay Samuel, Barry Wang, and Daphne Ippolito. NoveltyBench: Evaluating language models for human-like diversity. In *Conference on Language Modeling (COLM)*, 2025. arXiv:2504.05228.